

Initiating Search

November 10, 2025, 5:00 PM

• Search:

CAS Lexicon Search:

Chlorination - Preferred Concept

Filtered By:

Document Type: **Journal**Language: **English**

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon + Filters (12,614)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,125)	 Reactions	View Results
Filtered By:		
Yield: 90-100%, 80-89%, 70-79%		

Copyright © 2025 American Chemical Society (ACS). All Rights Reserved.

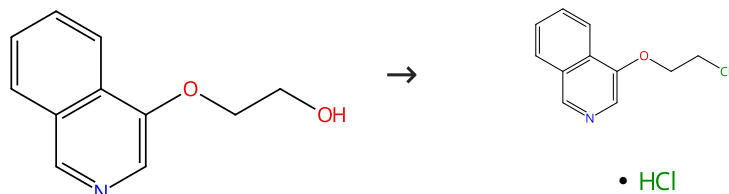
Internal use only. Redistribution is subject to the terms of your CAS SciFinder License Agreement and CAS information Use Policies.

Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (3)

31-048-CAS-16487287

Steps: 1 Yield: 100%

Acetylene-free synthesis of vinyloxy pyridine and quinoline

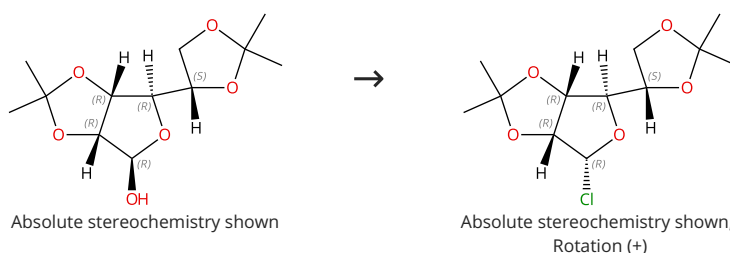
1.1 Reagents: Thionyl chloride
Solvents: Dichloromethane; 10 h, 60 °C

By: Hamdi, Abdelrahman; et al
Tetrahedron Letters (2016), 57(51), 5825-5829.

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-048-CAS-19425037

Steps: 1 Yield: 100%

A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions

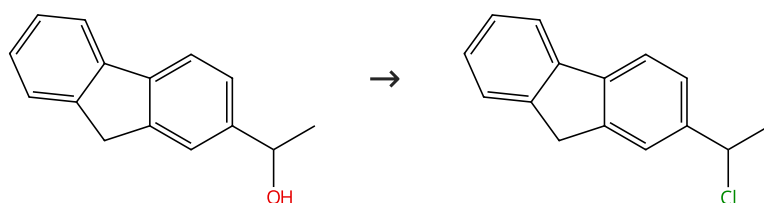
1.1 Reagents: Cyanuric chloride
Catalysts: 1-Pyrrolidinecarboxaldehyde
Solvents: 1,4-Dioxane; 8 h, 40 °C

By: Huy, Peter H.; et al
Chemistry - A European Journal (2018), 24(29), 7410-7416.

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (20)

 Suppliers (5)

31-048-CAS-12714173

Steps: 1 Yield: 100%

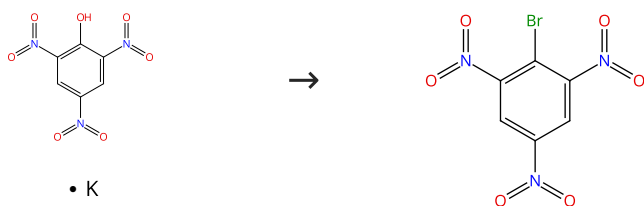
Chlorination of Benzylic and Allylic Alcohols with Trimethylsilyl Chloride Enhanced by Natural Sodium Montmorillonite

1.1 Reagents: Chlorotrimethylsilane
Catalysts: Montmorillonite ((Al_{1.33}-1.67Mg_{0.33}-0.67)(Ca_{0.1}Na_{0.1})
0.33Si₄(OH)₂O₁₀·xH₂O)
Solvents: Dichloromethane; 40 min, rt

By: Tandiary, Michael Andreas; et al
Synlett (2014), 25(18), 2639-2643.

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (7)

Suppliers (4)

31-048-CAS-9370333

Steps: 1 Yield: 100%

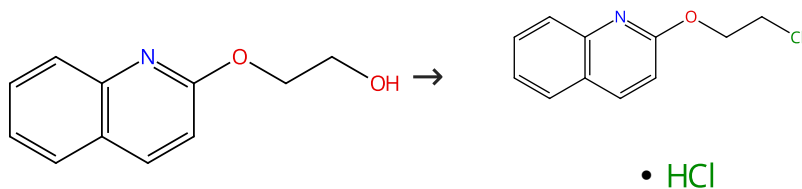
Chloro- and bromo-demethoxylation of methoxypolynitroaryl systems

1.1 Reagents: Phosphorus tribromide
Solvents: Dimethylformamide, Toluene

By: Bellamy, Anthony J.; et al
Tetrahedron (1996), 52(15), 5639-42.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (5)

31-048-CAS-16487291

Steps: 1 Yield: 100%

Acetylene-free synthesis of vinyloxy pyridine and quinoline

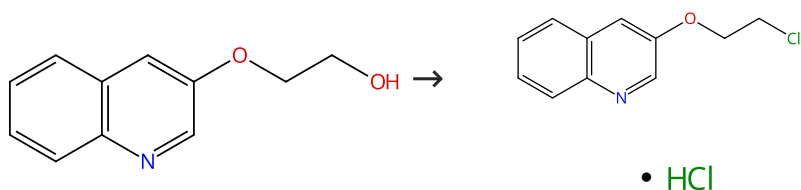
1.1 Reagents: Thionyl chloride
Solvents: Dichloromethane; 10 h, 60 °C

By: Hamdi, Abdelrahman; et al
Tetrahedron Letters (2016), 57(51), 5825-5829.

Experimental Protocols

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

31-048-CAS-16487286

Steps: 1 Yield: 100%

Acetylene-free synthesis of vinyloxy pyridine and quinoline

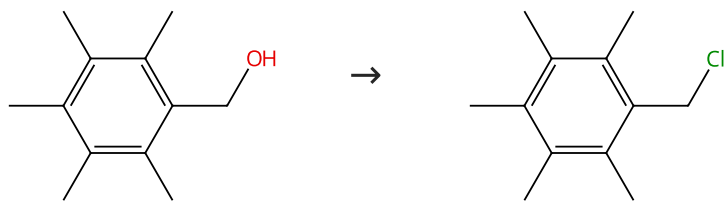
1.1 Reagents: Thionyl chloride
Solvents: Dichloromethane; 10 h, 60 °C

By: Hamdi, Abdelrahman; et al
Tetrahedron Letters (2016), 57(51), 5825-5829.

Experimental Protocols

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (46)

Suppliers (57)

31-048-CAS-11058695

Steps: 1 Yield: 100%

Chlorination of Benzylic and Allylic Alcohols with Trimethylsilyl Chloride Enhanced by Natural Sodium Montmorillonite

By: Tandiyar, Michael Andreas; et al

Synlett (2014), 25(18), 2639-2643.

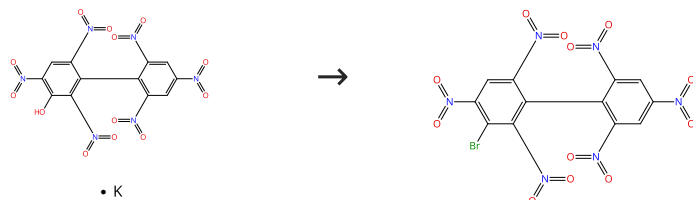
1.1 Reagents: Chlorotrimethylsilane

Catalysts: Montmorillonite ((Al_{1.33}-1.67Mg_{0.33}-0.67)(Ca_{0.1}Na_{0.1})_{0.33}Si₄(OH)₂O₁₀·xH₂O)

Solvents: Dichloromethane; 40 min, rt

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



31-048-CAS-13634535

Steps: 1 Yield: 100%

Chloro- and bromo-demethoxylation of methoxypolynitroaryl systems

By: Bellamy, Anthony J.; et al

Tetrahedron (1996), 52(15), 5639-42.

1.1 Reagents: Phosphorus tribromide

Solvents: Dimethylformamide, Toluene

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



31-048-CAS-10176346

Steps: 1 Yield: 100%

Structural Analysis of Cylindrical and Spherical Supramolecular Dendrimers Quantifies the Concept of Monodendron Shape Control by Generation Number

By: Percec, V.; et al

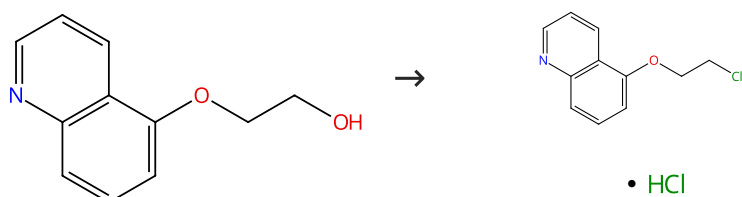
Journal of the American Chemical Society (1998), 120(43), 11061-11070.

1.1 Reagents: Thionyl chloride, 2,6-Di-*tert*-butyl-4-methylpyridine

Solvents: Dichloromethane; 20 °C

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (6)

31-048-CAS-16487288

Steps: 1 Yield: 100%

Acetylene-free synthesis of vinyloxy pyridine and quinoline

1.1 Reagents: Thionyl chloride
Solvents: Dichloromethane; 10 h, 60 °C

By: Hamdi, Abdelrahman; et al
Tetrahedron Letters (2016), 57(51), 5825-5829.

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (60)

Suppliers (64)

31-048-CAS-18636832

Steps: 1 Yield: 100%

Ir-Catalyzed Asymmetric and Regioselective Hydrogenation of Cyclic Allylsilanes and Generation of Quaternary Stereocenters via the Hosomi-Sakurai Allylation

1.1 Reagents: Thionyl chloride
Catalysts: Dimethylformamide
Solvents: Dichloromethane; 0 °C; 14 h, 0 °C

By: Rabten, Wangchuk; et al

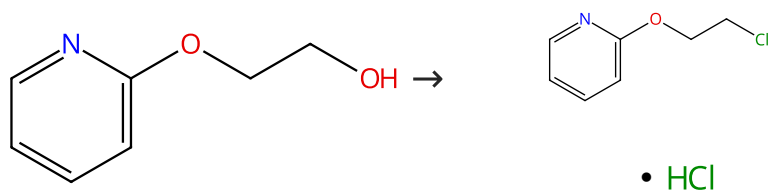
1.2 Reagents: Sodium bicarbonate
Solvents: Water

Chemistry - A European Journal (2018), 24(7), 1681-1685.

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (34)

31-048-CAS-16487290

Steps: 1 Yield: 100%

Acetylene-free synthesis of vinyloxy pyridine and quinoline

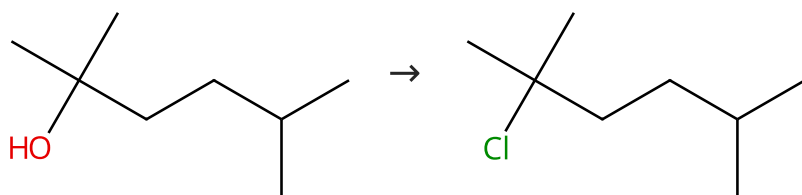
1.1 Reagents: Thionyl chloride
Solvents: Dichloromethane; 10 h, 60 °C

By: Hamdi, Abdelrahman; et al
Tetrahedron Letters (2016), 57(51), 5825-5829.

Experimental Protocols

Scheme 13 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (38)

Suppliers (32)

31-048-CAS-26750

Steps: 1 Yield: 100%

Direct chlorination of alcohols: synthesis of ethyl 3-chloro-3-phenylpropanoate [benzenepropanoic acid, β -chloro-, ethyl ester]

1.1 **Reagents:** Benzil, Chlorodimethylsilane
Catalysts: Indium trichloride
Solvents: Dichloromethane; 9 h, rt

By: Yasuda, Makoto; et al

Organic Syntheses (2006), 83, 38-44.

31-048-CAS-34645

Steps: 1 Yield: 100%

Indium-Catalyzed Direct Chlorination of Alcohols Using Chlorodimethylsilane-Benzil as a Selective and Mild System

1.1 **Reagents:** Benzil, Chlorodimethylsilane
Catalysts: Indium trichloride
Solvents: Dichloromethane; 9 h, rt

By: Yasuda, Makoto; et al

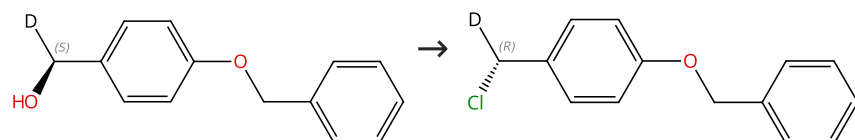
Journal of the American Chemical Society (2004), 126(23), 7186-7187.

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water; rt

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (+)Absolute stereochemistry shown,
Rotation (-)

31-048-CAS-13021263

Steps: 1 Yield: 100%

Chirally deuterated benzyl chlorides from benzyl alcohols via hexachloroacetone/polymer-supported triphenylphosphine: synthesis of protected (2S, 3S)-[3- 2 H, 15 N]-tyrosine

1.1 **Reagents:** Hexachloroacetone, Triphenylphosphine (polystyrene-supported)
Solvents: Tetrahydrofuran; 20 min, 0 °C

By: Barnett, Derek W.; et al

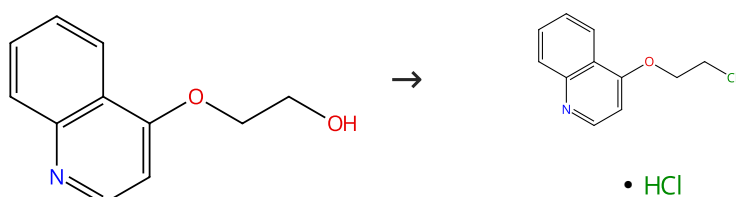
1.2 **Solvents:** Tetrahydrofuran; 0 °C; 1 h, 0 °C

Journal of Labelled Compounds and Radiopharmaceuticals (2013), 56(1), 6-11.

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%

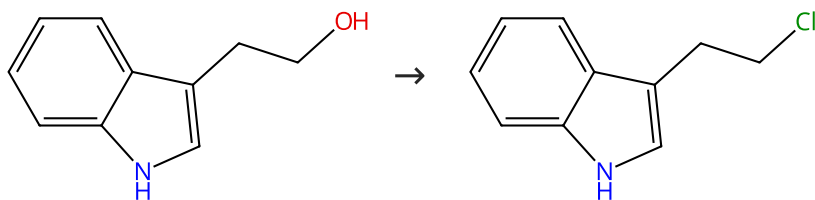


Suppliers (30)

31-048-CAS-16487284	Steps: 1 Yield: 100%	Acetylene-free synthesis of vinyloxy pyridine and quinoline
1.1 Reagents: Thionyl chloride Solvents: Dichloromethane; 10 h, 60 °C		By: Hamdi, Abdelrahman; et al
Experimental Protocols		Tetrahedron Letters (2016), 57(51), 5825-5829.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



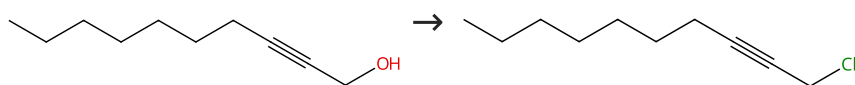
Suppliers (94)

Suppliers (27)

31-048-CAS-19425042	Steps: 1 Yield: 100%	A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions
1.1 Reagents: Cyanuric chloride Catalysts: 1-Pyrrolidinecarboxaldehyde Solvents: Acetonitrile; 2 h, 80 °C		By: Huy, Peter H.; et al
Experimental Protocols		Chemistry - A European Journal (2018), 24(29), 7410-7416.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



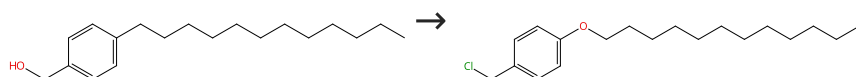
Suppliers (68)

Suppliers (3)

31-048-CAS-19425025	Steps: 1 Yield: 100%	A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions
1.1 Reagents: Cyanuric chloride Catalysts: 1-Pyrrolidinecarboxaldehyde Solvents: Ethyl acetate; 4 h, 40 °C		By: Huy, Peter H.; et al
Experimental Protocols		Chemistry - A European Journal (2018), 24(29), 7410-7416.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



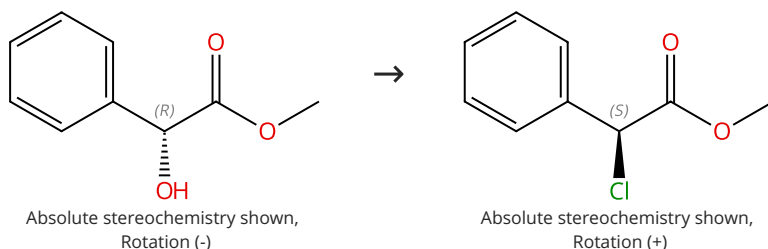
Suppliers (3)

Suppliers (18)

31-048-CAS-1169984	Steps: 1 Yield: 100%	Structural Analysis of Cylindrical and Spherical Supramolecular Dendrimers Quantifies the Concept of Monodendron Shape Control by Generation Number By: Percec, V.; et al Journal of the American Chemical Society (1998), 120(43), 11061-11070.
1.1 Reagents: Thionyl chloride Catalysts: Dimethylformamide Solvents: Dichloromethane; 20 °C		

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



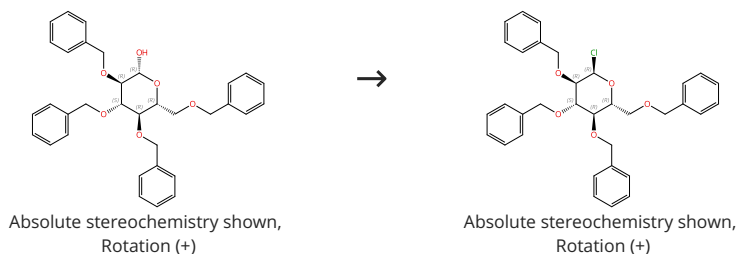
Suppliers (80)

Suppliers (3)

31-048-CAS-19425050	Steps: 1 Yield: 100%	A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions By: Huy, Peter H.; et al Chemistry - A European Journal (2018), 24(29), 7410-7416.
1.1 Reagents: Cyanuric chloride Catalysts: 1-Pyrrolidinecarboxaldehyde Solvents: 1,4-Dioxane; 20 h, 80 °C		
Experimental Protocols		

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



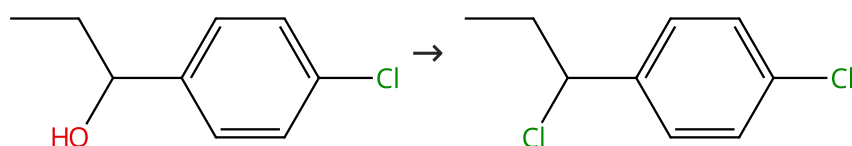
Suppliers (8)

Suppliers (28)

31-048-CAS-19425036	Steps: 1 Yield: 100%	A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions By: Huy, Peter H.; et al Chemistry - A European Journal (2018), 24(29), 7410-7416.
1.1 Reagents: Cyanuric chloride Catalysts: 1-Pyrrolidinecarboxaldehyde Solvents: 1,4-Dioxane; 24 h, 40 °C		
Experimental Protocols		

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



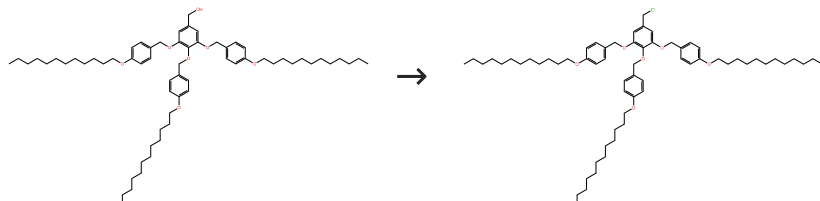
Suppliers (50)

Suppliers (6)

31-048-CAS-2701539	Steps: 1 Yield: 100%	A highly chemoselective and rapid chlorination of benzyl alcohols under neutral conditions
1.1 Reagents: Cyanuric chloride Solvents: Dimethyl sulfoxide; 25 min, rt		By: Sun, Lili; et al
Experimental Protocols		Synthesis (2008), (24), 3919-3924.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



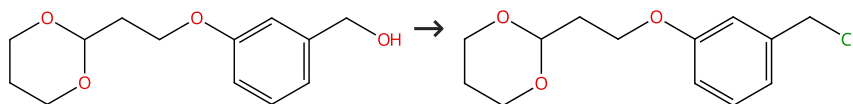
Suppliers (3)

Suppliers (11)

31-048-CAS-8425470	Steps: 1 Yield: 100%	Structural Analysis of Cylindrical and Spherical Supramolecular Dendrimers Quantifies the Concept of Monodendron Shape Control by Generation Number
1.1 Reagents: Thionyl chloride, 2,6-Di- <i>tert</i> -butyl-4-methylpyridine Solvents: Dichloromethane; 20 °C		By: Percec, V.; et al
		Journal of the American Chemical Society (1998), 120(43), 11061-11070.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%

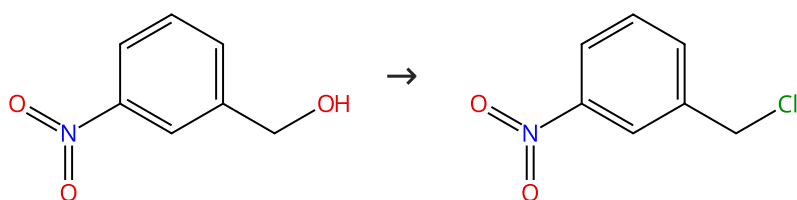


Suppliers (3)

31-048-CAS-19425024	Steps: 1 Yield: 100%	A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions
1.1 Reagents: Cyanuric chloride Catalysts: 1-Pyrrolidinecarboxaldehyde Solvents: Acetone; 4 h, 40 °C		By: Huy, Peter H.; et al
Experimental Protocols		Chemistry - A European Journal (2018), 24(29), 7410-7416.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



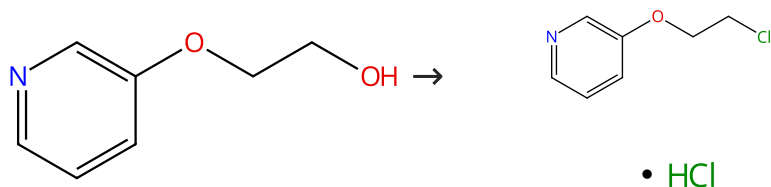
Suppliers (95)

Suppliers (51)

31-048-CAS-7282020	Steps: 1 Yield: 100%	A highly chemoselective and rapid chlorination of benzyl alcohols under neutral conditions
1.1 Reagents: Cyanuric chloride Solvents: Dimethyl sulfoxide; 15 min, rt		By: Sun, Lili; et al
Experimental Protocols		Synthesis (2008), (24), 3919-3924.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



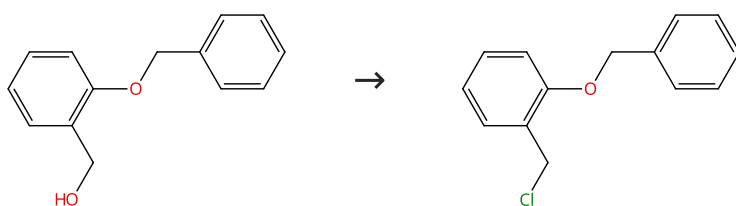
Suppliers (46)

Suppliers (3)

31-048-CAS-16487285	Steps: 1 Yield: 100%	Acetylene-free synthesis of vinyloxy pyridine and quinoline
1.1 Reagents: Thionyl chloride Solvents: Dichloromethane; 10 h, 60 °C		By: Hamdi, Abdelrahman; et al
Experimental Protocols		Tetrahedron Letters (2016), 57(51), 5825-5829.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



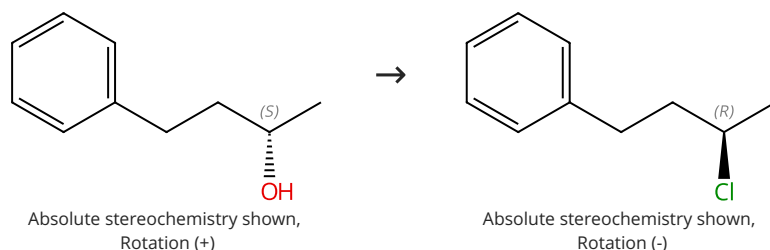
Suppliers (55)

Suppliers (50)

31-048-CAS-5123415	Steps: 1 Yield: 100%	A highly chemoselective and rapid chlorination of benzyl alcohols under neutral conditions
1.1 Reagents: Cyanuric chloride Solvents: Dimethyl sulfoxide; 10 min, rt		By: Sun, Lili; et al
Experimental Protocols		Synthesis (2008), (24), 3919-3924.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



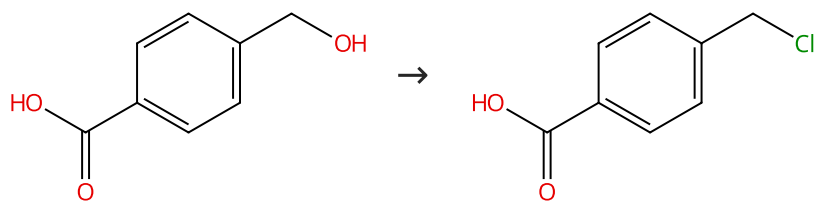
Suppliers (53)

Suppliers (3)

31-048-CAS-19425052	Steps: 1 Yield: 100%	A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions
1.1 Reagents: Cyanuric chloride Catalysts: 1-Pyrrolidinecarboxaldehyde Solvents: Ethyl acetate; 20 h, 80 °C		By: Huy, Peter H.; et al
Experimental Protocols		Chemistry - A European Journal (2018), 24(29), 7410-7416.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (104)

Suppliers (83)

31-048-CAS-19425029

Steps: 1 Yield: 100%

A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions

1.1 **Reagents:** Cyanuric chloride
Catalysts: 1-Pyrrolidinecarboxaldehyde
Solvents: 1,4-Dioxane; 13 h, 40 °C

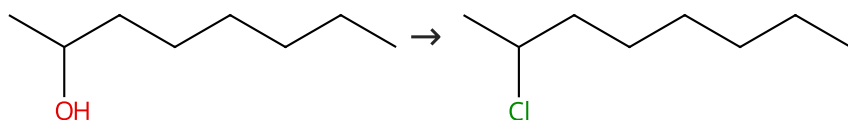
By: Huy, Peter H.; et al

Chemistry - A European Journal (2018), 24(29), 7410-7416.

Experimental Protocols

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (90)

Suppliers (16)

31-048-CAS-7094499

Steps: 1 Yield: 100%

An efficient method for chlorination of alcohols using PPh₃/Cl₃CCONH₂

1.1 **Reagents:** Trichloroacetamide, Triphenylphosphine
Solvents: Dichloromethane; 15 min, 30 °C

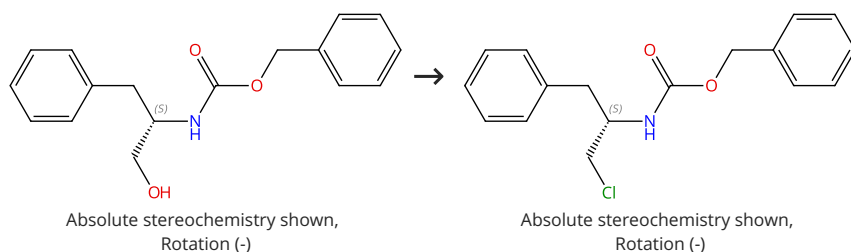
By: Pluempanupat, Wanchai; et al

Tetrahedron Letters (2006), 47(38), 6821-6823.

1.2 **Solvents:** Water

Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (72)

31-048-CAS-19425048

Steps: 1 Yield: 100%

A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions

1.1 **Reagents:** Cyanuric chloride
Catalysts: 1-Pyrrolidinecarboxaldehyde
Solvents: 1,4-Dioxane; 6 h, 80 °C

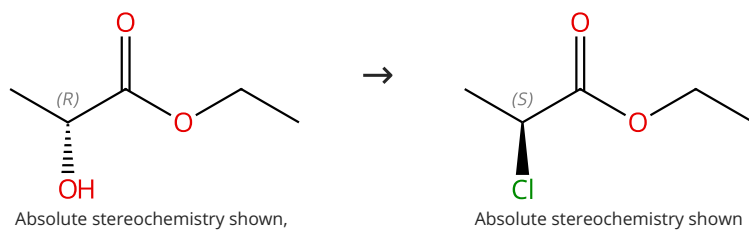
By: Huy, Peter H.; et al

Chemistry - A European Journal (2018), 24(29), 7410-7416.

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (59)

Suppliers (26)

31-048-CAS-4562502

Steps: 1 Yield: 100%

Optimization of the use of a chiral bio-based building block for the manufacture of DHPPA, a key intermediate for propionate herbicides

By: Fleer, Michel P. M.; et al

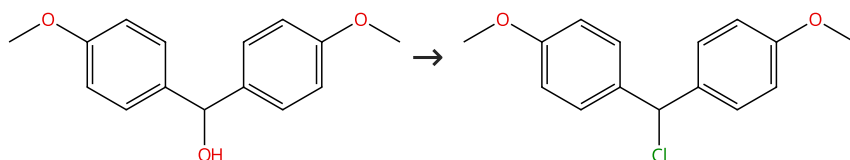
Green Chemistry (2014), 16(8), 3993-3998.

1.1 **Reagents:** Thionyl chloride
Catalysts: Dimethylformamide; rt → 65 °C; 65 °C; 65 °C → 87 °C; 4 h, 87 °C

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (71)

Suppliers (45)

31-048-CAS-6531218

Steps: 1 Yield: 100%

Chlorination of Benzylic and Allylic Alcohols with Trimethylsilyl Chloride Enhanced by Natural Sodium Montmorillonite

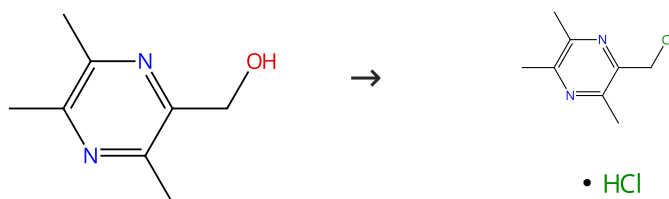
By: Tandiary, Michael Andreas; et al

Synlett (2014), 25(18), 2639-2643.

1.1 **Reagents:** Chlorotrimethylsilane
Catalysts: Montmorillonite ((Al_{1.33}-1.67Mg_{0.33}-0.67)(Ca_{0.1}Na_{0.1})
 0.33Si₄(OH)₂O₁₀·xH₂O)
Solvents: Dichloromethane; 40 min, rt

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (57)

Supplier (1)

31-048-CAS-15912324

Steps: 1 Yield: 100%

Ligustrazine Derivatives. Part 8: Design, Synthesis, and Preliminary Biological Evaluation of Novel Ligustrazinyl Amides as Cardiovascular Agents

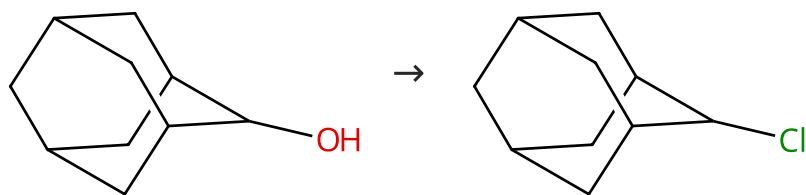
By: Li, Zhenyu; et al

Medicinal Chemistry (Sharjah, United Arab Emirates) (2014), 10(1), 81-89.

1.1 **Reagents:** Thionyl chloride
Solvents: Dichloromethane; 0 °C; 2.5 h, rt

Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (102)

Suppliers (51)

31-048-CAS-2244355

Steps: 1 Yield: 100%

An efficient method for chlorination of alcohols using $\text{PPh}_3/\text{Cl}_3\text{CONH}_2$ 1.1 **Reagents:** Trichloroacetamide, Triphenylphosphine**Solvents:** Dichloromethane; 15 min, 30 °C

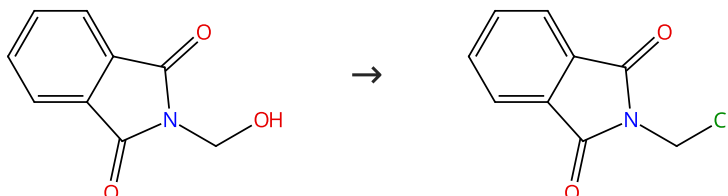
By: Pluemanupat, Wanchai; et al

1.2 **Solvents:** Water

Tetrahedron Letters (2006), 47(38), 6821-6823.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (82)

Suppliers (67)

31-048-CAS-10081647

Steps: 1 Yield: 100%

Chlorination of phosphorus pentasulfide

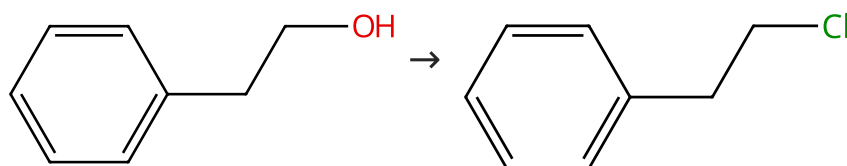
No Data Available

By: Zur, Zvi; et al

Chemistry & Industry (London, United Kingdom) (1975), (10), 436.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (110)

Suppliers (60)

31-048-CAS-2853779

Steps: 1 Yield: 100%

An efficient method for chlorination of alcohols using $\text{PPh}_3/\text{Cl}_3\text{CONH}_2$ 1.1 **Reagents:** Trichloroacetamide, Triphenylphosphine**Solvents:** Dichloromethane; 15 min, 30 °C

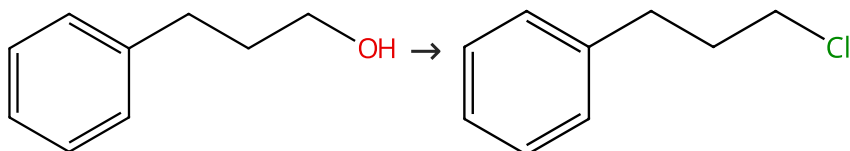
By: Pluemanupat, Wanchai; et al

1.2 **Solvents:** Water

Tetrahedron Letters (2006), 47(38), 6821-6823.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (67)

31-048-CAS-3174532

Steps: 1 Yield: 100%

A novel stereoselective one-pot conversion of alcohols into alkyl halides mediated by N,N'-diisopropylcarbodiimide

By: Crosignani, Stefano; et al

Chemical Communications (Cambridge, United Kingdom) (2003), (2), 260-261.

- 1.1 **Reagents:** Diisopropylcarbodiimide
Catalysts: Cuprous chloride
Solvents: Tetrahydrofuran; 5 min, 100 °C
- 1.2 **Reagents:** Acetyl chloride; 5 min, 150 °C

Scheme 38 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (365)

Suppliers (43)

31-048-CAS-19862

Steps: 1 Yield: 100%

Direct chlorination of alcohols: synthesis of ethyl 3-chloro-3-phenylpropanoate [benzenepropanoic acid, β-chloro-, ethyl ester]

By: Yasuda, Makoto; et al

Organic Syntheses (2006), 83, 38-44.

- 1.1 **Reagents:** Benzil, Chlorodimethylsilane
Catalysts: Indium trichloride
Solvents: Dichloromethane; 6 h, rt

31-048-CAS-31196

Steps: 1 Yield: 100%

Indium-Catalyzed Direct Chlorination of Alcohols Using Chlorodimethylsilane-Benzil as a Selective and Mild System

By: Yasuda, Makoto; et al

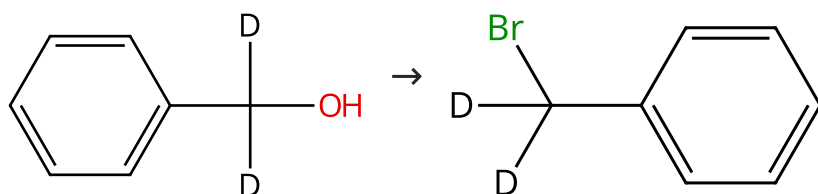
Journal of the American Chemical Society (2004), 126(23), 7186-7187.

- 1.1 **Reagents:** Benzil, Chlorodimethylsilane
Catalysts: Indium trichloride
Solvents: Dichloromethane; 6 h, rt
- 1.2 **Reagents:** Sodium bicarbonate
Solvents: Water; rt

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 100%



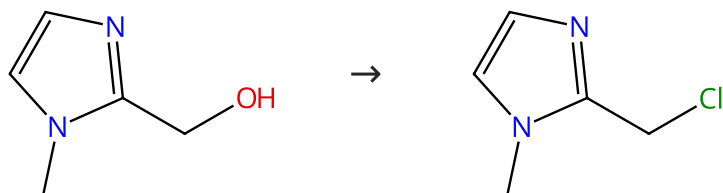
Suppliers (28)

Suppliers (23)

31-048-CAS-10762553	Steps: 1 Yield: 100%	Primary and secondary kinetic deuterium isotope effects and transition-state structures for benzylic chlorination and bromination of toluene By: Hanzlik, Robert P.; et al Journal of the American Chemical Society (1987), 109(16), 4926-30.
1.1 Reagents: Phosphorus tribromide Solvents: 1,2-Dichloroethane		

Scheme 40 (1 Reaction)

Steps: 1 Yield: 100%



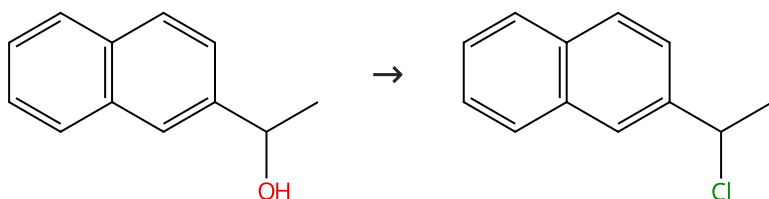
Suppliers (85)

Suppliers (55)

31-048-CAS-12417395	Steps: 1 Yield: 100%	Investigation of carbon-2 substituted imidazoles and their corresponding ionic liquids By: Liao, Chen; et al Tetrahedron Letters (2011), 52(41), 5308-5310.
1.1 Reagents: Thionyl chloride; 15 min, cooled; 15 min, reflux Experimental Protocols		

Scheme 41 (1 Reaction)

Steps: 1 Yield: 100%



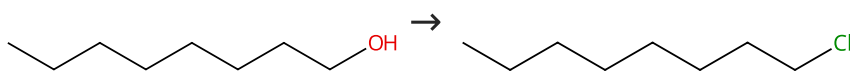
Suppliers (92)

Suppliers (10)

31-048-CAS-1220749	Steps: 1 Yield: 100%	Chlorination of Benzylic and Allylic Alcohols with Trimethylsilyl Chloride Enhanced by Natural Sodium Montmorillonite By: Tandiyar, Michael Andreas; et al Synlett (2014), 25(18), 2639-2643.
1.1 Reagents: Chlorotrimethylsilane Catalysts: Montmorillonite ((Al _{1.33} -1.67Mg _{0.33} -0.67)(Ca ₀₋₁ Na ₀₋₁) _{0.33} Si ₄ (OH) ₂ O ₁₀ ·xH ₂ O) Solvents: Dichloromethane; 40 min, rt		

Scheme 42 (2 Reactions)

Steps: 1 Yield: 100%



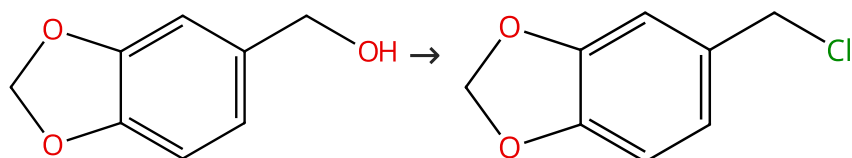
Suppliers (132)

Suppliers (56)

31-048-CAS-7385682 Steps: 1 Yield: 100% 1.1 Reagents: Triphenylphosphine, (7-4)-Dichlorodiphenyl selenium Solvents: Benzene	Triphenylphosphine/Dichloroselenurane: A new reagent for a selective conversion of alcohols into alkyl chlorides By: Drabowicz, Jozef; et al Journal of Organic Chemistry (1998), 63(25), 9565-9568.
31-048-CAS-5078755 Steps: 1 Yield: 100% 1.1 Reagents: Triphenylphosphine, Selenium, dichlorodimethyl-, (7-4)- Solvents: Chloroform	Triphenylphosphine/Dichloroselenurane: A new reagent for a selective conversion of alcohols into alkyl chlorides By: Drabowicz, Jozef; et al Journal of Organic Chemistry (1998), 63(25), 9565-9568.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%



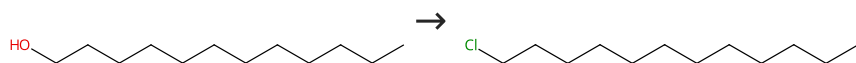
Suppliers (80)

Suppliers (54)

31-048-CAS-3037710 Steps: 1 Yield: 100% 1.1 Reagents: Cyanuric chloride Solvents: Dimethyl sulfoxide; 20 min, rt Experimental Protocols	A highly chemoselective and rapid chlorination of benzyl alcohols under neutral conditions By: Sun, Lili; et al Synthesis (2008), (24), 3919-3924.
---	---

Scheme 44 (1 Reaction)

Steps: 1 Yield: 100%



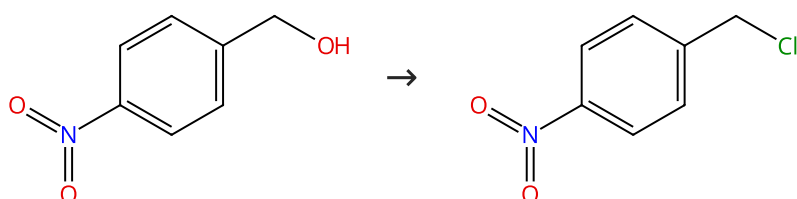
Suppliers (99)

Suppliers (66)

31-048-CAS-19425038 Steps: 1 Yield: 100% 1.1 Reagents: Cyanuric chloride Catalysts: 1-Pyrrolidinecarboxaldehyde Solvents: Acetonitrile; 13 h, 80 °C Experimental Protocols	A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions By: Huy, Peter H.; et al Chemistry - A European Journal (2018), 24(29), 7410-7416.
--	---

Scheme 45 (1 Reaction)

Steps: 1 Yield: 100%



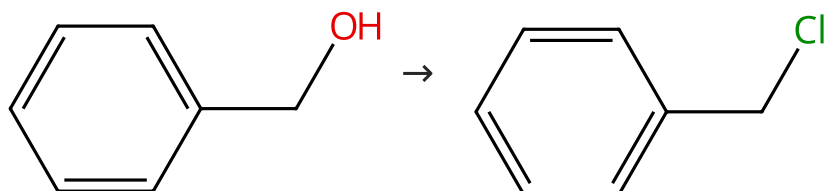
Suppliers (96)

Suppliers (68)

31-048-CAS-19425030	Steps: 1 Yield: 100%	A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions
1.1 Reagents: Cyanuric chloride Catalysts: 1-Pyrrolidinecarboxaldehyde Solvents: Acetone; 20.5 h, 40 °C		By: Huy, Peter H.; et al
Experimental Protocols		Chemistry - A European Journal (2018), 24(29), 7410-7416.

Scheme 46 (1 Reaction)

Steps: 1 Yield: 100%



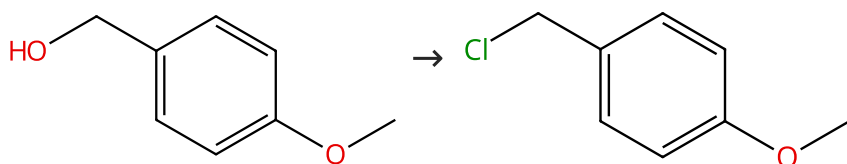
Suppliers (149)

Suppliers (65)

31-048-CAS-2376302	Steps: 1 Yield: 100%	Preparation of aryl chlorides from phenols
1.1 Reagents: Chlorine Solvents: Chloroform		By: Bay, Elliott; et al
1.2 Reagents: Dichlorophenylphosphine		Journal of Organic Chemistry (1990), 55(10), 3415-17.

Scheme 47 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (105)

Suppliers (70)

31-048-CAS-19425031	Steps: 1 Yield: 100%	A General Catalytic Method for Highly Cost- and Atom-Efficient Nucleophilic Substitutions
1.1 Reagents: Cyanuric chloride Catalysts: 1-Pyrrolidinecarboxaldehyde Solvents: Ethyl acetate; 6 h, 40 °C		By: Huy, Peter H.; et al
Experimental Protocols		Chemistry - A European Journal (2018), 24(29), 7410-7416.